

1 Propagations of Uncertainty

Relates the uncertainty of a calculated result, r , to the uncertainties in the input variables

Two approaches will be used:

1. Kline-McClintock(KM) -Analytical approaches
2. Monte Carlo (MC) -Purely computational approach (Preferred)

1.1 Kline-McClintock

Suppose r is a function of L random variables:

$$r = f(x_1, x_2, x_3, x_L)$$

Then an uncertainty interval for r is:

$$r = \bar{r} \pm k_r u_r (P\%)$$

$$k_r \equiv \text{coverage factor} = \lim_{v \rightarrow \infty} t_{v,p}$$

$$\bar{r} = f(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_L)$$

$$u_r = \sum_{i=1}^L ((\phi_{x_i} u_{x_i})^2)^{1/2} = \text{standard uncertainty in } r \text{ (sum of the square root of the variance)}$$

$$\phi_{x_i} = \left. \frac{df}{dx_i} \right|_{x_i=\bar{x}_i} = \text{sensitivity index}$$

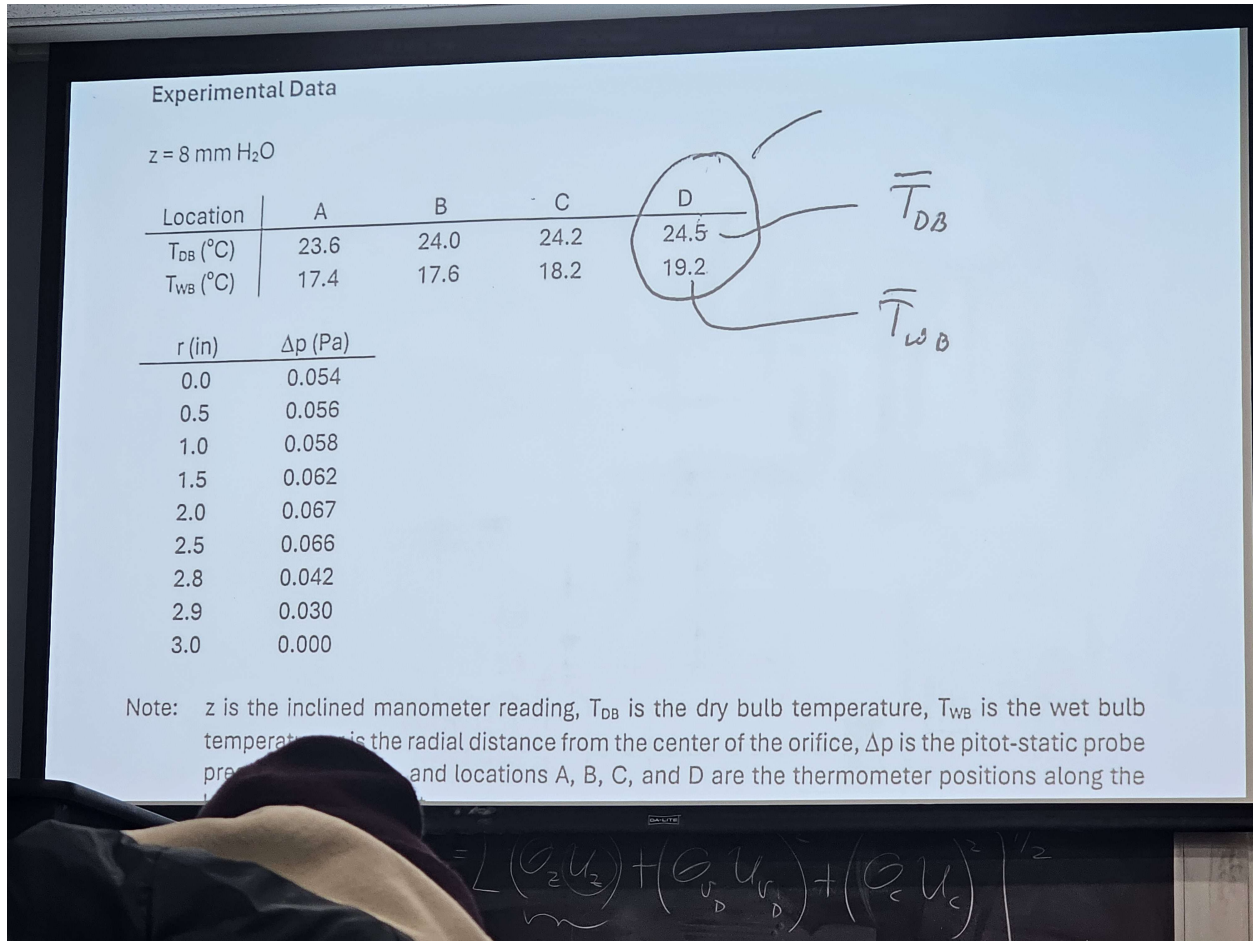
1.1.1 Check with other notes

For the project, need to calculate $\dot{m}$ UI's(uncertainty interval) for three cases

1. Manufacturer's correlation
2. Psychrometric chart reading error
3. Obstruction meter theory

(Maybe he was just talking about the three random variables affecting mass flow in the next section)

1.1.2 Example



from the general form:

$$r = f(x_1, x_2, x_3)$$

$$u_r = [(\phi_{x_1} u_{x_1})^2 + (\phi_{x_2} u_{x_2})^2 + (\phi_{x_3} u_{x_3})^2]^{1/2}$$

when applied to mass flow rate:

$$\dot{m} = f(Z, v_d, C)$$

$$u_{\dot{m}} = [(\phi_z u_z)^2 + (\phi_{v_d} u_{v_d})^2 + (\phi_c u_c)^2]^{1/2}$$

Where:

$$\phi_{x_1} = \frac{df}{dx_i} = \frac{df}{dz} = \frac{d}{dz} [cz^{1/2} v_d^{-1/2}] = \phi_z$$

$$\phi_z = \left. \frac{1}{2} cz^{-1/2} v_d^{-1/2} \right|_{z=\bar{z}, v_d=\bar{v}_d, c=\bar{c}} = \frac{1}{2\bar{c}} \frac{1}{\bar{z}^{1/2} \bar{v}_d^{1/2}}$$

$$r = r(x_1, x_2, x_3)$$

$$u_r = \left[\left(\theta_{x_1} u_{x_1} \right)^2 + \left(\theta_{x_2} u_{x_2} \right)^2 + \left(\theta_{x_3} u_{x_3} \right)^2 \right]^{1/2}$$

$$u_m = \left[\left(\theta_z u_z \right)^2 + \left(\theta_{v_D} u_{v_D} \right)^2 + \left(\theta_c u_c \right)^2 \right]^{1/2}$$

$$\bar{c} = 0.0504$$

$$\bar{v}_D = 0.858 \frac{\text{m}^3}{\text{kg}}$$

$$\bar{z} = 8 \text{ mm H}_2\text{O}$$

$\bar{v}_D$ = Value from the psychrometric chart (or calculated)

$$\dot{m} = c z^{1/2} v_D^{-1/2}$$

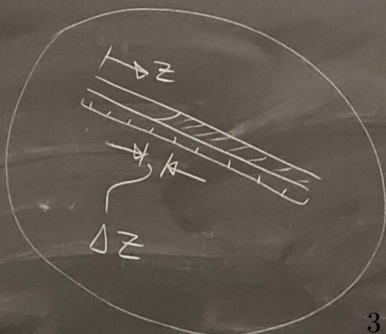
$$\theta_{v_D} = \frac{\partial \dot{m}}{\partial v_D} = -\frac{c}{z} z^{1/2} v_D^{-3/2}$$

$$\theta_{v_D} = -\frac{\bar{c}}{\bar{z}} \bar{z}^{1/2} \bar{v}_D^{-3/2}$$

$$\theta_z = \frac{1}{z} (0.0504) \frac{1}{8^{1/2} (0.858)^{1/2}} = 0.0096 \frac{\text{kg}}{\text{m}^3 \text{ mm H}_2\text{O}}$$

$$u_z = \left(s_z^2 + b_z^2 \right)^{1/2}$$

$$u_{v_D} = ?$$



2 Psychrometric chart reading error

3 Obstruction meter theory

$$Q_1 = K_0 A_0 \left(\frac{2\Delta P}{\rho} \right)^{1/2} Y$$

$$\dot{m} = \rho_1 Q_1 = K_0 A_0 (2\Delta P \rho_1)^{1/2} Y$$

$$\dot{m} = f(K_0, A_0, \rho_1, \Delta P, Y)$$